BDA400 | DATA SCIENCE TOOLS AND TECHNIQUES

Technical Analysis Using R

Assignment 5 | Development Stage

Student	Belal Al-bahrazawi	**Portfolio**	AAPL, MSFT, NVDA
Repository	github.com/belalalbahrazawi111-boop/TechnicalAnalysis	**Language**	R (base functions for indicators)

Project Overview

This project implements nine technical analysis indicators in R, then combines them in a small daily stock-data workflow. It reads one CSV per portfolio symbol, applies the selected price source and timeframe, calculates indicators, applies an example trading rule, and writes a result CSV for each symbol.

Indicator Implementations

Indicator	Function and implementation summary
SMA	sma(data, period) calculates each complete rolling window; output length is n - period + 1.
EMA	ema(data, period) seeds from the first value and applies the 2/(period+1) recurrence.
MACD	macd(...) returns the MACD line, signal line, and histogram from the EMA components.
Standard deviation	stdev(data) calculates population standard deviation using denominator n.
Linear regression	linreg(...) computes slope, intercept, and fitted values for the assignment's selected window.
RSI	rsi(data, period) uses initial average gains/losses and Wilder smoothing.
StochRSI	stoch_rsi(...) normalizes the valid RSI series, then smooths %K and %D with SMA.
Crossover	crossover(arr1, arr2) reports Up, Down, or None for each relation transition.
Crossunder	crossunder(arr1, arr2) returns TRUE when arr1 crosses below arr2 from at or above.

Integration and Verification

The runner reads the symbols from `Assignment_5/portfolio.txt` and their daily CSV files from `Assignment_5/data`. It supports daily, weekly, or monthly bars, an inclusive date range, and Adjusted or Close as the price source.

Example Trading Rule

The demonstration labels a row Buy when EMA(12) crosses above EMA(26) while RSI(14) is below 70. It labels Sell when EMA(12) crosses below EMA(26) or RSI(14) is above 70; other rows are Hold. This is an instructional example, not investment advice.

RStudio Verification

> **All nine indicator check groups passed in RStudio on November 6, 2026.**

The test output confirmed SMA windows, EMA recurrence, MACD components and histogram identity, population standard deviation, regression values, RSI behavior, StochRSI range, crossover and crossunder events, and invalid-input checks. The daily pipeline then completed and saved indicator CSV files for AAPL, MSFT, and NVDA in `Assignment_5/outputs`.

Run Instructions

From the repository root in RStudio, run:

```
source("Assignment_5/scripts/test_a05.R")
run_a05_tests()
source("Assignment_5/scripts/run_a05.R")
run_a05(timeframe = "daily", price_source = "Adjusted")
```

Implementation Notes

The indicator calculations use base R and preserve the function names and parameters from the supplied templates. SMA returns compact rolling-window results; the integration workflow aligns those values with dates. StochRSI follows the assignment pseudocode, which normalizes across the valid RSI series before SMA smoothing. The assignment's crossover and crossunder descriptions differ; this implementation follows the specified Up/Down/None output for crossover and TRUE/FALSE events for crossunder.

Before submission, review the code and generated outputs, and verify that the Assignment_5 folder and this report are present in the GitHub repository.

RStudio Execution Evidence

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The console shows all indicator checks passing and the daily output CSVs saved for AAPL, MSFT, and NVDA.

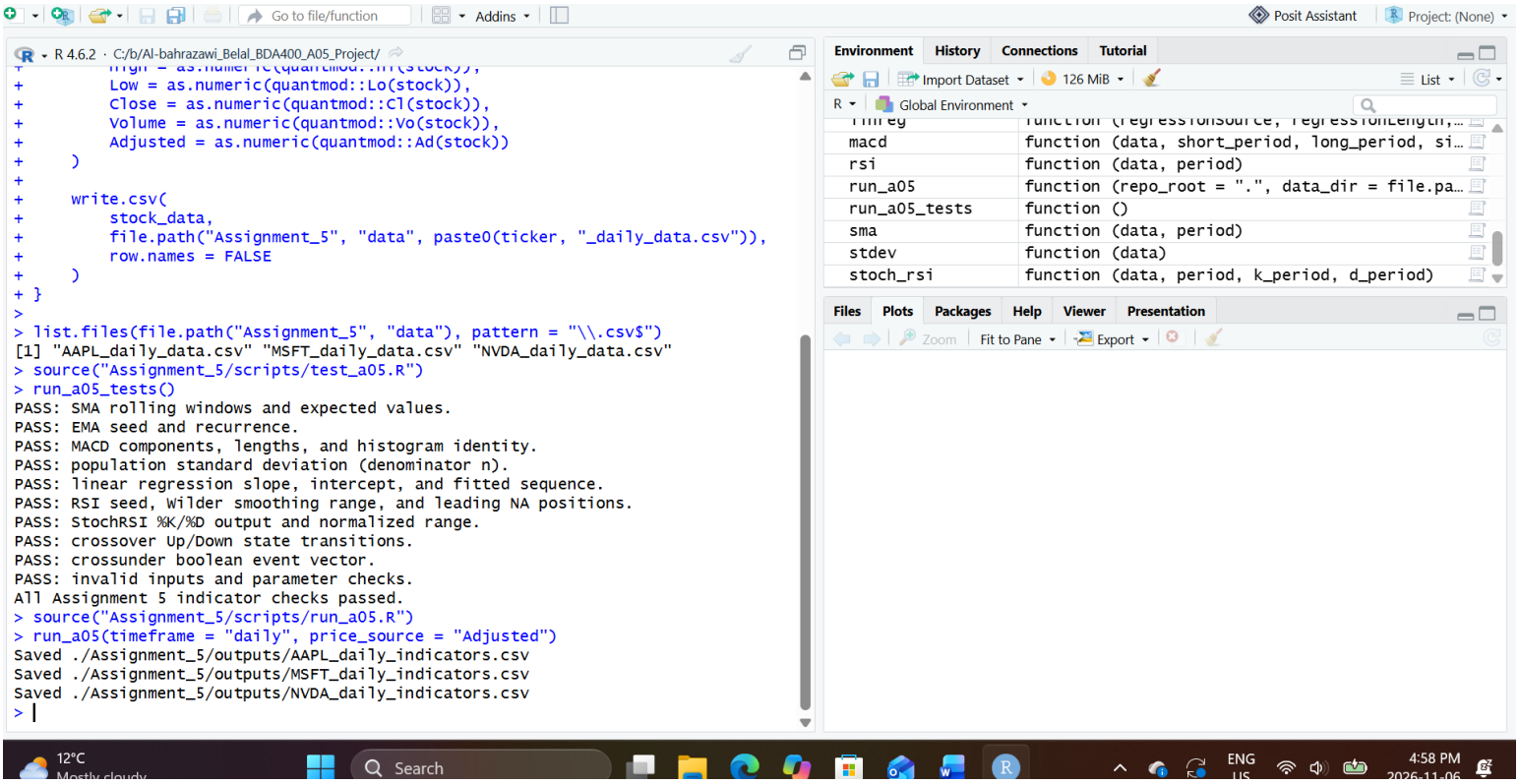


Figure 1. RStudio test results and successful daily pipeline run.